

Territory Mapping Pipeline

What is built, how it works, and how it reaches the app

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Region: **bengaluru_jayanagar_lalbagh** (Jayanagar → Lalbagh corridor, Bengaluru)

1. Why this pipeline exists

The game is played on **real places** — parks, lakes, named neighbourhood blocks — not arbitrary grids. Milestone 1 proved the run loop with ~8 hand-drawn territories. Milestone 2 replaces hand-drawing with a **deterministic, human-reviewed** build pipeline that turns OpenStreetMap and Overture data into exhaustive territory polygons for an area of interest (AOI).

Core product rule: the map is not the product; the feeling of changing the map is. Territories must read as places a local would recognise.

2. End-to-end architecture

Layer	Component	Role
Sources	Overture Maps + Geofabrik OSM	Pinned downloads; checksums in manifest
Build	tools/pipeline/ (stages 01–09)	Offline GIS; never runs on the phone
Review	staging/candidates.geojson	Human opens in QGIS/geojson.io; APPROVED file
Publish	published/territories.geojson	Immutable audit artifact + manifest
Serve	FastAPI + PostGIS	GET /v1/territories for the mobile app
Client	Expo + MapLibre	Renders polygons; server matches runs

Pipeline flow (build time):

01 download → 02 separators + 03 landmarks → 04 boundary graph → 05 polygonize & carve → 06 normalize sizes → 07 assign names → 08 validate (fail closed) → [human APPROVED] → 09 publish

3. Pipeline stages (01–09)

#	Stage	Summary
01	download_data	Overture + Geofabrik extracts, SHA-256, manifest
02	extract_features	Road/rail separators from highway_classes.yaml
03	extract_landmarks	Named places; role major minor; only majors protected
04	build_boundary_graph	Snap, node, drop dangles; working CRS (UTM 43N)
05	polygonize_and_carve	Faces inside AOI; carve protected majors whole
06	normalize_sizes	Merge scraps; freeze protected; safe geometry
07	assign_names	Place → landmark → street-pair → review; stable territory_id
08	validate	Hard topology gate; candidates only on pass
09	publish	Copy candidates to published/; gated on APPROVED

Key invariants

- Every point in the AOI belongs to exactly one territory (no gaps, no overlaps).
- All measurement in metres (UTM 43N); storage in EPSG:4326.
- Protected landmarks (e.g. Lalbagh) are never merged, split, or resized.
- territory_id = uuid5(city/area/slug) — permanent; geometry changes bump version only.
- Stage 08 fails closed: invalid topology → report only, no candidates file.
- No AI naming; names come from OSM/Overture or deterministic fallbacks.

4. Where files live

All pipeline output under **data/pipeline/bengaluru/jayanagar_lalbagh/**:

Path	Contents
raw/	Download manifest + source extracts
intermediate/	02_separators ... 07_named.geojson
staging/	candidates.geojson, validation_report.json, chain_fingerprint.json
staging/APPROVED	Human-created marker (not written by pipeline)
published/	territories.geojson, publish_manifest.json

5. What is built today

Pipeline: All nine stages implemented and tested.

Published AOI (jayanagar_lalbagh): 2357 territories (29 protected landmarks), validation pass.

Typical fabric median area: ~8,888 m² (target soft band 20,000–80,000 m² for anonymous street fabric).

Mobile + backend (Milestone 1 loop):

- Expo app with MapLibre (custom dev client; not Expo Go).
- Foreground GPS recording; SQLite run durability.
- FastAPI backend: territories GeoJSON, ownership state, run upload + matching.
- Placeholder ownership rule: highest cumulative distance wins.
- Pipeline publish can be seeded into PostGIS via backend/scripts/seed_territories.py.

6. Sizing & thresholds (config/thresholds.yaml)

Key	Value	Meaning
scrap_m2	5,000	Below this: merge into neighbour (scrap)
fabric_soft_min_m2	20,000	Target lower bound for anonymous fabric
fabric_soft_max_m2	80,000	Target upper bound (~1–2 territories per run)
territory_min_area_m2	20,000	Floor for protected major landmarks
validation_min_coverage	0.999	Stage 08 hard gate

Playtest note: Current fabric is finer than the soft band (many small grid cells). Coarser merges or place-based dissolve are the likely next tuning pass — not random shapes.

7. How to run the pipeline

```
cd tools/pipeline && uv sync
uv run python run_pipeline.py --list
uv run python run_pipeline.py --region bengaluru_jayanagar_lalbagh
uv run python run_pipeline.py --region bengaluru_jayanagar_lalbagh --from-stage 6 --to-stage 8
# After human review:
touch data/pipeline/bengaluru/jayanagar_lalbagh/staging/APPROVED
uv run python run_pipeline.py --region bengaluru_jayanagar_lalbagh --from-stage 9
# Seed mobile API:
cd backend && uv run python scripts/seed_territories.py --source pipeline
```

8. Deferred (not in this milestone)

- PostGIS upsert from Stage 09 (publish is file copy today; seed script bridges to API).
- data/territories/ versioned export layout for backend seed (partial bridge exists).
- H3 territory_cells indexing.
- Influence model, decay, clubs, anti-cheat detection.
- AI-assisted naming.
- Admin console; file-based APPROVED gate only.

Canonical contract: docs/09_TERRITORY_GENERATION_PIPELINE.md · Operational detail: tools/pipeline/README.md